

Section Criteria for Integral Lorentzian Lattices and the Cost of Realization

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Abstract

For an integral lattice Λ_W of signature (3,1) and a positive definite integral L of rank 3 we give a **criterion** for L to be a **primitive timelike section** of Λ_W by a primitive vector of norm $-n$. The criterion runs on three data — **signature, bilinear discriminant form and parity** (type I/II) — and states: such a section exists **if and only if** there is an anti-isometry of glue subgroups $\gamma : H_L \rightarrow H_t$ whose graph Γ satisfies $\Gamma^\perp/\Gamma \cong \text{disc}(\Lambda_W)$ **as an isometry of forms**, and whose glued overlattice has **the same parity** as Λ_W . Both requirements are load-bearing: the order-only version of the first is insufficient, and without the second the statement is **false** — a counterexample on the ambient $\text{diag}(1, 1, 2, -2)$ is exhibited explicitly in §3.4. The criterion is stated **relative to a fixed ambient** and is expressed **exclusively** in discriminant data and parity — **the Gram matrix does not enter it**. The reverse implication rests on two checkable conditions: **(T0)** ("the named data determine the genus of the ambient") and **(T0c)** ("the genus of the ambient contains one class"); for rank 4, signature (3,1) and $|\det| \leq 4$ — a scope covering all four ambients of this work — both are **proved**: (T0) by a complete enumeration of genera (there are exactly seven, and the triple of data separates them pairwise, §3.5), (T0c) by Eichler's classical theorem with a margin of 5^6 (§3, Remark 1).

Two direct checks. **First**, substituting $\text{disc}(\Lambda_W) = 0$ returns the condition $\langle 1/n \rangle$ of the unimodular case, that is, the theorem of preprint-6: the generalisation **collapses back onto what is already in print**. **Second**, the parity rider shows that this boundary is **sharp**: for $p \equiv 1(\text{mod}8)$ the condition $\langle 1/n \rangle$ is no longer sufficient, because two lattices with an **identical** discriminant form fall into different ambients, and what separates them is one extra bit — parity. The case $p = 3$ is thereby isolated ($II_{3,1}$ does not exist), so the printed result of preprint-6 **stands as it is**.

From the criterion it follows that the **minimal discriminant of an ambient** — the "cost" of a lattice L — depends **only on the bilinear discriminant form** of L , and is therefore an **invariant of its genus**; we give for it a **closed expression** $\text{cost}(L) = \min|\text{disc}(L)| \cdot n_{\min}(m)/m^2 : m|\exp(\text{disc}L)$. Finiteness of the range here is **structural**, not computational, so the expression contains no search bound — and that is exactly why it is **robust against truncation**, which in earlier bounded computations three times produced inflated values. Evaluating the expression on fourteen representatives gives a table of costs; it is indexed by the **discriminant form**, not by Bravais type and not by the order of the discriminant, because cost is **not** an invariant of the type and is **not monotone** in the order. In particular, the centrings of the cubic lattice acquire exact

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addresses: **FCC** — **3**, **BCC** — **8**, while the blocking of lower values for FCC splits into **two different** mechanisms (insolubility of $k^2 \equiv 3(mod4)$ at $d = 1$; insolubility of $2k^2 \equiv 3(mod4)$ at $d = 2$), whereas $d = 3$ is blocked by nothing at all.

There is no physical reading anywhere in this work: the signature $(3,1)$ appears as an arithmetic condition — in particular as the **condition of applicability** of the classical theorem on the genus of indefinite forms, on which the criterion leans in one explicitly named step.

§1 · Statement of the problem

Preprint-6 (Zenodo 10.5281/zenodo.22068307) classified the primitive timelike sections of the **unimodular** lattice $I_{p,1}$ by the condition $\langle 1/n \rangle$ (P6.Thm1(7)).

The question of this work: what replaces that condition for an **arbitrary** integral Lorentzian lattice.

What exactly is being asked, and under which quantifier: the lattice Λ_W is taken as **GIVEN**. The whole work answers the question

"is L a primitive timelike section of **THIS** lattice Λ_W ",

and **not** "does there exist **some** ambient into which L embeds". The difference is not rhetorical: under the second reading the answer is almost always "yes" and the question is empty; it acquires content **only** for a fixed ambient — and it is precisely this quantifier that the statement of **P7.Thm1** carries. The one place where the ambient is **not** fixed but **sought** is the definition of **cost** (P7.Def5): there the minimum is taken over all admissible Λ_W , and this is said in the definition itself.

Scope, stated once and at the start: this work contains **no physical reading whatsoever**. No object is read as "time", "energy" or "mass"; the signature $(3,1)$ appears **exclusively** as an arithmetic condition — and in one place (Remark 1) as the **condition of applicability** of a classical theorem.

§2 · Definitions

P7.Def1 — discriminant form. For a non-degenerate integral L : $disc(L) := L^\vee/L$ with the **bilinear** form $b_L : disc(L) \times disc(L) \rightarrow \mathbb{Q}/\mathbb{Z}$. The abbreviation b is used where the discriminant form is meant as an **invariant** (with no lattice named in the subscript).

Why $bmod\mathbb{Z}$ and not $qmod2\mathbb{Z}$: the quadratic version is defined only for **even** lattices, while $\langle -n \rangle$ is **odd for odd n** — and that is not an edge case but the main one. The whole criterion is therefore stated in the bilinear form. The price of this choice is named explicitly: the bilinear form **does not see parity**, so parity enters the data of the criterion as a **separate bit** (§3).

P7.Def2 — glue subgroups and anti-isometry. Subgroups $H_L \subseteq disc(L)$, $H_t \subseteq disc(\langle -n \rangle)$ are called **glue subgroups**. An isomorphism $\gamma : H_L \rightarrow H_t$ is an **anti-isometry** if $b_t(\gamma x, \gamma y) = -b_L(x, y)$.

The term is not invented: "glue" is the established word of the construction ([cited] Conway–Sloane, SPLAG ch. 4: *glue group*, *glue vectors*; Nikulin §1.5). We name only the **subgroups** on which the gluing is defined, because all the corollaries refer to them.

P7.Def3 — the graph. $\Gamma := (x, \gamma x) : x \in H_L \subseteq \text{disc}(L) \oplus \text{disc}(\langle -n \rangle)$. Γ is **isotropic** if $b_L \oplus b_t$ vanishes on Γ — for the graph of an anti-isometry this holds **identically**: $b_L(x, y) + b_t(\gamma x, \gamma y) = b_L(x, y) - b_L(x, y) = 0$.

Two sides of one equivalence: here, "anti-isometry $\implies$ isotropy"; in (Thm1.3), the converse. Together: for subgroups with injective projections, **isotropic graphs are exactly the graphs of anti-isometries.**

P7.Def4 — section. An integral positive definite L of rank 3 (abstract) is a **primitive timelike section** of Λ_W by t if $t \in \Lambda_W$ is **primitive** ($t/k \notin \Lambda_W$ for every integer $k \geq 2$), $\langle t, t \rangle = -n < 0$, and there is an isometric embedding $\iota : L \hookrightarrow \Lambda_W$ with $\iota(L) = t^\perp \cap \Lambda_W$. The shorthand $L = t^\perp \cap \Lambda_W$ is henceforth read through this ι — the reverse implication proves the **existence of an embedding**, not an equality of sets.

Primitivity is required of t — and this requirement is load-bearing; for L it is automatic. Automatic: $\Lambda_W \cap (L \otimes \mathbb{Q}) = \Lambda_W \cap t^\perp = L$ by definition — the intersection of a lattice with a subspace is always primitive. Load-bearing: without the requirement on t the theorem of §3 is **false**. Witness: $\Lambda_W = I_{3,1}$, $t = 2e_4$, $n = 4$, $L = t^\perp \cap \Lambda_W = \mathbb{Z}^3$ — the section exists, but the right-hand leg of the criterion fails: $\text{disc}(L) = 0 \implies$ the only anti-isometry is the empty one, $\Gamma = 0$, and $\Gamma^\perp/\Gamma \cong \mathbb{Z}/4 \not\cong \text{disc}(I_{3,1}) = 0$. The same $\mathbb{Z}^3$ is a section by the primitive e_4 with $n = 1$ — the norm of a section is well defined only on a primitive vector.

P7.Def5 — cost.

$\text{cost}(L) := \min \{ |\det \Lambda_W| : \Lambda_W \text{ integral, of signature } (3,1), \text{ and } L \text{ a primitive timelike section of } \Lambda_W \text{ by some vector } \}$

P7.Def6 — $j_{\min}$ and the admissible glue orders.

$D(L) := \{ m \geq 1 : m \mid \exp(\text{disc } L) \}$
 $j_{\min}(m) := \min \{ j \geq 1 : \exists x \in \text{disc}(L) \text{ of order } m, \exists \text{ a unit } k \pmod m, j \cdot k^2 \equiv b_L(x, x) \cdot m \pmod m \} , m \in D(L)$
 $n_{\min}(m) := m \cdot j_{\min}(m)$

Why $\exp$ and not $|\text{disc}|$: taking $m \mid |\text{disc}(L)|$ would admit an m with no element of order m , i.e. the term under $\min$ would be **undefined**. Example: $\mathbb{Z}/2 \times \mathbb{Z}/2$ has $4 \mid 4$ but $\exp = 2$. For the computation this changes nothing — **for the statement it changes everything**. It is a separate definition because $j_{\min}$ stands **in the statement** of the theorem and is referred to by the corollaries: a quantity named only inside a theorem gets cited by transcription.

§3 · P7.Thm1 — the gluing criterion

P7.Thm1. Let L be a positive definite integral lattice of rank 3, $n \geq 1$, and let Λ_W be an integral lattice of signature (3,1) (**both fixed**) satisfying the conditions **(T0)** the triple of data (**signature, bilinear discriminant form, parity**) determines the genus of Λ_W uniquely among integral lattices — a condition **checkable by a finite comparison** (for rank 4, $\text{sig } (3,1)$, $|\det| \leq 4$ — in particular for all four ambients

of this work — it is **proved** by an enumeration of genera, P7.Prop1, §3.5); **(T0c)** the genus of Λ_W contains **one isometry class**. (Class = spinor genus for indefinite lattices of rank ≥ 3 — [cited] Eichler; a genus with more than one spinor genus in rank 4 requires $|\det| \geq 5^6 = 15625$ — [cited] SPLAG Ch. 15, Cor. 22; hence for $|\det| \leq 4$ condition (T0c) holds with room to spare — Remark 1, leg 1. Without (T0c) the reverse leg yields only " Λ lies in the genus of Λ_W ", not an isometry.) Then L is a **primitive timelike section of Λ_W by a primitive vector of norm $-n$** (P7.Def4) $\iff$ **there exists an anti-isometry of glue subgroups $\gamma : H_L \rightarrow H_t$** whose graph Γ satisfies **both** conditions: **(T1)** $\Gamma^\perp/\Gamma \cong \text{disc}(\Lambda_W)$ — **as an isometry of forms**, not merely as an isomorphism of groups; **(T2)** the glued overlattice $\Lambda(\Gamma)$ has **the same parity** (type I/II) as Λ_W ; parity is read off the norms of the glue vectors $\text{mod } 2\mathbb{Z}$. (Isotropy of Γ is automatic by P7.Def3.)

(T1) "as a form", not "as a group". The order-only version fails at the first non-cyclic discriminant (**P7.Cor3**, §6): $\mathbb{Z}/4$ and $\mathbb{Z}/2 \times \mathbb{Z}/2$ have the same order 4 and are not isometric. But the **group** alone is not enough either: on $\mathbb{Z}/6$ there live **two non-isometric forms**, and they give **different costs** — b on the generator equal to $5/6$ gives 2, and $1/6$ gives 1 (§7.2 against the row oP in §7.4). A group is not a form.

(T2) parity is the third datum, and it is not decorative. The criterion runs on the triple **signature + bilinear discriminant form + parity**. With these data the equivalence holds for **all four** ambients of this work: for three of them an even competitor **does not exist** (excluded by Milgram's formula — table **§3.4**), and for the fourth the **bit separates** the odd $\text{diag}(1, 1, 2, -2)$ from the even $A_1^2 \oplus U$. Without **(T2)** the statement is **false** precisely on the fourth — the explicit pair is in §3.4.

Remark 1 — what the theorem stands on (to be read BEFORE the proof)

The step requires the data of the criterion to determine Λ_W **uniquely**, not merely up to genus. It stands on **two** legs, and they must be seen separately — their conflation was the reason the second went unchecked for a long time.

Leg 1 — "class = genus" for our objects. The classical theorem (**Eichler; Kneser**) for **indefinite** integral lattices of **rank ≥ 3** states: **class = SPINOR genus**. [cited] a named import, **not re-derived** in this work. Our ambient has rank 4, so the rank condition is met. The classical theorem gives "**class = spinor genus**", whereas the step needs "**class = genus**". These are **not identical**: a genus may contain **several** spinor genera. For our objects the difference is void, and this is a count, not an estimate: in rank 4 a genus with **more than one** spinor genus requires $|\det| \geq 5^6 = 15625$ ([cited] SPLAG Ch. 15, Cor. 22); our four ambients have $|\det| \in 1, 2, 4, 4$ — a margin of almost four thousandfold, so the condition holds **trivially**, not "at the edge". This is exactly condition **(T0c)** of the theorem: within the scope of this work it is proved; for an arbitrary Λ_W it stands in the statement as a separate hypothesis (a genus with several classes would make the conclusion of (Thm1.11) false).

Leg 2 — parity, and the limit of what it gives. "Signature + **bilinear** discriminant form" **do not determine the genus**: parity is a genus invariant that the bilinear form **does not see**. Counterexample: $A_1^3 \oplus \langle -2 \rangle$ glues by different Γ into the **even** $A_1^2 \oplus U$ and the **odd** $\text{diag}(1, 1, 2, -2)$ — same signature, same discriminant group and b , different genera. That is exactly why parity stands in the data of the criterion as the separate condition **(T2)**.

The limit of leg 2 — and the scope in which it is closed. From the fact that parity is **necessary** it does not follow that the triple "signature + b + parity" is **sufficient** to determine the genus of an **arbitrary** lattice: the 2-adic genus symbol

of an odd lattice is in general finer than a single global bit (it carries the type and the oddity of each Jordan constituent). Sufficiency therefore stands as the separate condition **(T0)** — and for rank 4, sig (3,1), $|\det| \leq 4$ it is **proved** by a complete enumeration of genera (**P7.Prop1**, §3.5): there are exactly seven genera, seven triples, and the correspondence is one-to-one. For arbitrary det, (T0) is **not claimed** — this work gives no general characterisation of (T0)-lattices and does not need one.

Without these two legs the criterion yields only: Λ lies in **the same genus** as Λ_W — that is, "a section of **some** lattice of the genus" instead of "a section of Λ_W **itself**". The whole force of the " $\Longleftrightarrow$ " rests on this step.

Counterfactual — why signature (3,1) is a condition of applicability. For definite lattices, class and genus **may differ**, and the argument of (Thm1.11) is **unavailable** there; hence for a definite ambient the criterion in the stated form would be unjustified. Signature (3,1) is a **condition of applicability**, not a convenience of notation.

P7.Lem1 — the gluing identity (bilinear, with no parity restriction)

P7.Lem1. Let M be a non-degenerate integral lattice, $\Lambda \supseteq M$ an integral overlattice of finite index, $\Gamma := \Lambda/M \subseteq M^\vee/M = \text{disc}(M)$, and $\Gamma^\perp := \{x \in \text{disc}(M) : b(x, \Gamma) = 0 \text{ in } \mathbb{Q}/\mathbb{Z}\}$. Then $\text{disc}(\Lambda) \cong \Gamma^\perp/\Gamma$ — as groups with a bilinear form $\text{mod } \mathbb{Z}$.

Proof. $M \subseteq \Lambda$ (given); integrality of Λ gives $\Lambda \subseteq \Lambda^\vee$; from $M \subseteq \Lambda$ we get $\Lambda^\vee \subseteq M^\vee$ — hence the chain $M \subseteq \Lambda \subseteq \Lambda^\vee \subseteq M^\vee$. For $x \in M^\vee$, the condition $b(x, \Lambda) \subseteq \mathbb{Z}$ is equivalent to $b(x, \gamma) \in \mathbb{Z}$ for all $\gamma \in \Gamma$ (on M integrality is automatic), i.e. $(x \text{ mod } M) \in \Gamma^\perp$; hence $\Lambda^\vee/M = \Gamma^\perp$. Since $\Lambda/M = \Gamma$, the third isomorphism theorem gives $\text{disc}(\Lambda) = \Lambda^\vee/\Lambda = (\Lambda^\vee/M)/(\Lambda/M) = \Gamma^\perp/\Gamma$, and the bilinear form on $\Lambda^\vee/\Lambda$ is induced from that of M identically. ■

Why a lemma here rather than a citation. The classical exposition of this identity (Nikulin) runs through the **quadratic** discriminant form $q \text{ mod } 2\mathbb{Z}$, defined only for **even** lattices — whereas our case is odd. The group-theoretic and bilinear side of the identity, as the proof shows, **needs no parity at all** — so for odd lattices it is proved rather than cited. A numerical check of both sides (by independent computations) was carried out on three odd gluings and one even control.

P7.Lem2 — gluing by an isotropic subgroup

P7.Lem2. Let M be a non-degenerate integral lattice, $\Gamma \subseteq \text{disc}(M)$ a subgroup, and $\Lambda_\Gamma := \{x \in M^\vee : x \text{ mod } M \in \Gamma\}$. Then: **(a)** Λ_Γ is a subgroup of $M^\vee$ with $M \subseteq \Lambda_\Gamma$ and $\Lambda_\Gamma/M = \Gamma$; Λ_Γ is **integral** $\Longleftrightarrow \Gamma$ is **isotropic**; **(b)** for $M = L \oplus \langle t \rangle$: L is primitive in $\Lambda_\Gamma \Longleftrightarrow$ the projection $\Gamma \rightarrow \text{disc}(\langle t \rangle)$ is **injective**; symmetrically, $\langle t \rangle$ is primitive in $\Lambda_\Gamma \Longleftrightarrow$ the projection $\Gamma \rightarrow \text{disc}(L)$ is **injective**.

Proof. **(a)** Λ_Γ is the preimage of the subgroup Γ under $M^\vee \rightarrow \text{disc}(M)$, hence a subgroup; M is the preimage of zero, so $M \subseteq \Lambda_\Gamma$ and $\Lambda_\Gamma/M = \Gamma$. Integrality: for $x, y \in \Lambda_\Gamma$ the value $b(x, y) \text{ mod } \mathbb{Z}$ equals $b(x, \bar{y})$, hence $b(\Lambda_\Gamma, \Lambda_\Gamma) \subseteq \mathbb{Z} \Longleftrightarrow b$ vanishes on Γ (pairs with $x = 0$ are integral automatically: $x \in M, y \in M^\vee$). **(b)** $\Lambda_\Gamma \cap (L \otimes \mathbb{Q}) \subseteq M^\vee \cap (L \otimes \mathbb{Q}) = L^\vee$, so an element of the intersection has image of the shape $(x, 0) \in \Gamma$; hence $\Lambda_\Gamma \cap (L \otimes \mathbb{Q}) = L \Longleftrightarrow \Gamma$ contains no non-zero element with vanishing $\langle t \rangle$ -component $\Longleftrightarrow \Gamma \rightarrow \text{disc}(\langle t \rangle)$ is injective. Symmetrically for $\langle t \rangle$. ■

The lemma closes in one place the four steps that lean on it: (Thm1.2)/(Thm1.3) in the forward leg (there $\Lambda_W = \Lambda_\Gamma$ for $\Gamma = \Lambda_W/M$), (Thm1.7)/(Thm1.10) in the reverse leg, and the chain (Thm2.9).

§3.1 · Proof: the forward leg (section $\implies \gamma$)

(**Thm1.1**) t primitive, $\langle t, t \rangle = -n$, $L = t^\perp \cap \Lambda_W$ (primitive automatically, P7.Def4) $\implies L \oplus \langle t \rangle \subseteq \Lambda_W$ — a sublattice of finite index. (**Thm1.2**) $\Gamma := \Lambda_W / (L \oplus \langle t \rangle)$; integrality of $\Lambda_W \implies \Gamma$ is **isotropic** (P7.Lem2(a), the direction "integral $\implies$ isotropic"; for the graph of an anti-isometry this also holds identically by P7.Def3). (**Thm1.3**) Primitivity of L (automatic) gives injectivity of $\Gamma \rightarrow \text{disc}(\langle t \rangle)$; primitivity of $\langle t \rangle$ (the requirement of P7.Def4 — exactly here is where it bears) gives injectivity of $\Gamma \rightarrow \text{disc}(L)$ (both — P7.Lem2(b)) $\implies \Gamma$ is the graph of an isomorphism γ . The anti-isometry is read **bilinearly**, from isotropy on a pair of distinct elements $(x, \gamma x)$, $(y, \gamma y) \in \Gamma$:

$$0 = (\mathbf{b}_L \oplus \mathbf{b}_t)((x, \gamma x), (y, \gamma y)) = \mathbf{b}_L(x, y) + \mathbf{b}_t(\gamma x, \gamma y) \implies \mathbf{b}_t(\gamma x, \gamma y) = -\mathbf{b}_L(x, y)$$

for **all** x, y — exactly what P7.Def2 demands. (It cannot be derived through the diagonal $b(x, x)$: over $\mathbb{Q}/\mathbb{Z}$ polarisation does not recover a bilinear form on 2-torsion.) (**Thm1.4**) $\text{disc}(\Lambda_W) \cong \Gamma^\perp / \Gamma$ as a form — condition (**T1**). (P7.Lem1.) (**Thm1.4'**) The norms of the glue vectors of $\Lambda_W / (L \oplus \langle t \rangle) \bmod 2\mathbb{Z}$ give the parity of Λ_W — condition (**T2**) holds identically, since the overlattice is Λ_W itself. (**Thm1.5**) $\text{disc}(\langle t \rangle) \cong \mathbb{Z}/n$ with $b(k, k) = -k^2/n$. ■

§3.2 · Proof: the reverse leg ($\gamma \implies \Lambda_W$)

(**Thm1.6**) $M := L \oplus \langle -n \rangle$, signature $(3, 0) + (0, 1) = (3, 1)$. (**Thm1.7**) An isotropic $\Gamma \subseteq \text{disc}(M)$ determines an **integral** overlattice $\Lambda := \Lambda_\Gamma \supseteq M$, $\Lambda/M \cong \Gamma$ (P7.Lem2(a)). (**Thm1.8**) The signature is preserved $\implies \Lambda$ has $(3, 1)$. (**Thm1.9**) $\text{disc}(\Lambda) \cong \Gamma^\perp / \Gamma$ (P7.Lem1 — the same lemma as in (Thm1.4)), and $\Gamma^\perp / \Gamma \cong \text{disc}(\Lambda_W)$ — by condition (**T1**). (**Thm1.10**) Primitivity — **two different properties, each with its own ground** (P7.Lem2(b)): L primitive $\iff$ injectivity of $\Gamma \rightarrow \text{disc}(\langle t \rangle)$; $\langle t \rangle$ primitive $\iff$ injectivity of $\Gamma \rightarrow \text{disc}(L)$; both injectivities are automatic, since Γ is the graph of an **isomorphism** γ . (**Thm1.11**) Λ and Λ_W have the same signature, discriminant form **and parity** (the last by condition (**T2**)). By condition (**T0**) these data determine the genus of Λ_W uniquely $\implies \Lambda$ and Λ_W lie in one genus; by condition (**T0c**) that genus contains one class $\implies \Lambda \cong \Lambda_W$. For the ambients of this work both conditions are proved: (T0) by the enumeration of genera (P7.Prop1, §3.5), (T0c) by [cited] **Eichler** plus the 5⁶ margin (Remark 1, leg 1). ■

§3.3 · The measurement that accompanies the proof

Two **independent routes** to one number: **route (1)** — a direct enumeration of primitive t with $t^2 = -n \leq 4$; **route (2)** — the gluing formula $|\text{disc}(L)| \cdot n = |\text{disc}(\Lambda_W)| \cdot \text{index}^2$.

ambient	'\{'\}	disc\{'\}	'	what the enumeration produced	(1) (2)	against
$I_{3,1}$ (control, disc = 0)	1	'n $\rightarrow$ '\{'\}	disc(L)\{'\}	= n, cyclic $\mathbb{Z}/n$ '	100 %	

ambient	$\backslash \{ \}$	disc $\backslash \{ \}$	‘	what the enumeration produced	(1) (2)	against
diag(1, 1, 1, -2)	2	1 $\rightarrow$ 2, 2 $\rightarrow$ 1, $\mathbb{Z}/2 \times \mathbb{Z}/2$,	100 %			
		3 $\rightarrow$ 6,				
		4 $\rightarrow \mathbb{Z}/2 \times \mathbb{Z}/4$				
diag(1, 1, 1, -4)	4	1 $\rightarrow$ 4, 2 $\rightarrow$ $\mathbb{Z}/2 \times \mathbb{Z}/4$, 3 $\rightarrow$ 12, 4 $\rightarrow$ 1	100 %			
$A_3 \oplus \langle -1 \rangle$	— 4	1 $\rightarrow$ 4 (= A_3), 2 $\rightarrow \mathbb{Z}/2 \times \mathbb{Z}/4$, 3 $\rightarrow$ 12	100 %			
the same ambient as the row above						

The last two rows describe **one** lattice: $\text{diag}(1, 1, 1, -4) \cong A_3 \oplus \langle -1 \rangle$ — both have a vector of norm -1 , in both the complement is A_3 , the theta series are identical. The identity of their columns is not a coincidence but a consequence of this isometry. $\implies$ **the measurement covered three distinct ambients out of four.** The fourth, $\text{diag}(1, 1, 2, -2)$, enters the work as the witness of §6 and §3.4; the test "(1) against (2)" could not have caught anything on it in any case — both routes enumerate **actual** sections, so a false reverse leg is invisible to them. That is exactly why a counterexample (§3.4) was needed, and not a wider search.

The unimodular row is a **control, not an illustration**: the enumeration reproduced $\langle 1/n \rangle$ by itself, without being tuned to it. This is **§4 in action**.

§3.4 · When (T2) holds by itself: exclusion of the even competitor

Condition **(T2)** is non-trivial only where an **even competitor exists** — that is, where there is an even lattice with the same signature and the same b as Λ_W . This question is settled by Milgram’s **necessary** condition, and the answer for the four ambients of this work fits into a table.

The mechanism. An even lattice carries a quadratic refinement $q : \text{disc} \rightarrow \mathbb{Q}/2\mathbb{Z}$ with $b_q = b$. Milgram’s formula ([cited] §10) relates the signature of the lattice to the Gauss sum of the refinement:

$$\text{GS}(q) = \sum_x e^{\pi i \cdot q(x)} = \sqrt{|\text{disc}|} \cdot e^{\pi i \cdot \sigma(q)/4} , \quad \sigma(\Lambda) \equiv \sigma(q) \pmod{8}$$

Signature (3,1) gives $\sigma = 2$. Hence: if **no** refinement q of the form b has $\sigma(q) \equiv 2 \pmod{8}$, then an even lattice with these data **does not exist** — there is no competitor, and (T2) holds automatically. There are finitely many refinements (on a cyclic generator of order m the value $q(g)$ is determined by $b(g, g)$ up to $\mathbb{Z}/2\mathbb{Z}$), so the enumeration is **complete**, not selective.

ambient	disc	$\sigma(q)$ — all attainable	even competitor
$I_{3,1}$	trivial	0	excluded
diag(1, 1, 1, -2)	$\mathbb{Z}/2$	1, 7	excluded
diag(1, 1, 1, -4)	$\mathbb{Z}/4$	3, 7	excluded
diag(1, 1, 2, -2)	$\mathbb{Z}/2 \times \mathbb{Z}/2$	0, 2, 6	NOT excluded

Three conclusions.

(i) On the first three, $2 \notin \sigma(q) \implies$ there is no even competitor $\implies$ **(T2) holds by itself**, and there the criterion works on the signature and b alone. (ii) The first row is **Remark 2** (§5) in wider notation: for a trivial discriminant the only refinement $q = 0$ gives $\sigma(q) = 0$, i.e. "an even unimodular lattice of signature (3,1) does not exist". Two statements that looked different are **one mechanism**, $\sigma \bmod 8$. (iii) On the fourth, $\sigma(q) = 2$ is **attainable** — and a competitor is indeed there: $A_1^2 \oplus U$, constructed explicitly below. So the table **does not over-squeeze**: it does not exclude what exists. It is on this row that (T2) carries its weight.

The explicit pair of the fourth row. $L = A_1^3$ (Gram $\text{diag}(2, 2, 2)$), $n = 2$, $\Lambda_W = \text{diag}(1, 1, 2, -2)$: condition **(T1)** holds — $H_L = \langle g_3 \rangle$, $\gamma : g_3 \mapsto 1$ is an anti-isometry ($\frac{1}{2} = -\frac{1}{2} \bmod \mathbb{Z}$), and $\Gamma^\perp / \Gamma \cong (\mathbb{Z}/2)^2$ with $b = \text{diag}(\frac{1}{2}, \frac{1}{2})$, which is **isometric** to $\text{disc}(\text{diag}(1, 1, 2, -2))$. But the gluing yields $\Lambda = A_1^2 \oplus U$ — the glue vector $w = (e_3 + t)/2$ has $w^2 = 0$, and the pair $w, (e_3 - t)/2$ is hyperbolic $\implies \Lambda$ is **even**, whereas Λ_W is **odd**. Condition **(T2)** is precisely what filters this case out.

On the status of this table. Milgram gives a **necessary** condition: $2 \notin \sigma(q)$ proves the **non-existence** of a competitor strictly. The converse — that for $\sigma(q) = 2$ a competitor does exist — does **not** follow from Milgram, and for the fourth row it is proved **not** by a Gauss sum but by the **explicit construction** of $A_1^2 \oplus U$. Two different methods, each used where it proves something.

§3.5 · P7.Prop1 — condition (T0) proved by an enumeration of genera

P7.Prop1. Integral lattices of rank 4, signature (3,1), $|\det| \in 1, 2, 4$ exist, up to genus, in **exactly seven** kinds, and the triple **(discriminant group, b , parity)** takes on them **seven distinct values**. Hence the triple determines the genus: condition **(T0)** holds for **every** lattice of this scope — in particular for all four ambients of this work.

Proof. The genus of an indefinite lattice is determined by the signature and the p-adic symbols; for $|\det| \in 1, 2, 4$ only the 2-adic one is non-trivial. The enumeration is run from above — from the existence conditions — and is closed from below by explicit lattices.

(1) The enumeration is complete. The existence conditions for a system of p-adic symbols ([cited] SPLAG Ch. 15 §7.7: the determinant condition, the oddity formula, the conditions on each Jordan constituent; Theorem 11 — such a system is realised by an integral form) give, for signature (3,1) and $|\det| \in 1, 2, 4$, **17 putative systems**. The merging of equivalent systems — *oddity fusion* and *sign walking* — is carried out **per Allcock–Gal–Mark** ([cited]: Lemma 5.2 — fusion only when the image is a legal symbol; Lemma 6.1 — the three admissible cases of a sign walk; **Theorem 6.2 — two symbols are isometric $\iff$ related by a chain of these operations**), so the count is exact **in both directions**: the operations neither merge inequivalent systems nor leave equivalent ones unmerged:

'\{'	$\det \backslash \{'$	'	putative tems	sys-	canonical era	gen-
1	1	1				
2	4	1				
4	12	5				
total	17	7				

Without canonicalisation the count is wrong: $|\det| = 2$ would read as "four genera" instead of one.

Why the merging follows AGM rather than the "canonical form" of SPLAG §7.6. The uniqueness claim of §7.6 ("at most one minus sign per train") is **false** in general — the error is named by Allcock–Gal–Mark (§6, following Allcock, *Memoirs AMS* 220); the correction is *signways*. Step (1) therefore rests not on §7.6 but on the AGM operations with Theorem 6.2, and its correctness does not hang on the absence of a known bad configuration: **all 17 systems are merged exclusively by operations licensed by Lemma 5.2 and Lemma 6.1** — no forbidden sign walk occurs in any chain — and Theorem 6.2 identifies the classes of this equivalence with the isometry classes, of which there are exactly $1 + 1 + 5 = 7$. A measured control: an unconditional train walk (without the conditions of Lemma 6.1) yields **eight** "genera" instead of seven — the count is sensitive to the rule, so the agreement with step (3) is not a tautology of the instrument.

(2) The seven genera by name.

$\backslash \{ \}$	$\det \backslash \{ \}$	$\cdot$	canonical 2-adic symbol	discriminant parity group	representative
1	$1_I^{+4}(t = 2)$	trivial	odd	$I_{3,1}$	
2	$1_I^{+3}(t = 1) \cdot 2_I^{+1}(t = 1)$	$\mathbb{Z}/2$	odd	$\text{diag}(1, 1, 1, -2)$	
4	$1_I^{+3}(t = 1) \cdot 4_I^{+1}(t = 1)$	$\mathbb{Z}/4$	odd	$\text{diag}(1, 1, 4, -1),$ $b = \frac{1}{4}$	
4	$1_I^{+3}(t = 3) \cdot 4_I^{+1}(t = 7)$	$\mathbb{Z}/4$	odd	$\text{diag}(1, 1, 1, -4) \cong$ $A_3 \oplus \langle -1 \rangle,$ $b = \frac{3}{4}$	
4	$1_I^{+2}(t = 0) \cdot 2_I^{+2}(t = 2)$	$\mathbb{Z}/2 \times \mathbb{Z}/2$	odd	$\text{diag}(1, 1, 2, -2)$	
4	$1_I^{+2}(t = 2) \cdot 2_{II}^{+2}(t = 0)$	$\mathbb{Z}/2 \times \mathbb{Z}/2$	odd	$I_2 \oplus U(2),$ b with zero di- agonal	
4	$1_{II}^{+2}(t = 0) \cdot 2_I^{+2}(t = 2)$	$\mathbb{Z}/2 \times \mathbb{Z}/2$	even	$A_1^2 \oplus U$	

A homonym is named: t inside the symbols is the oddity of a Jordan constituent (SPLAG §7.4 notation); it has nothing in common with the timelike vector t of the criterion (P7.Def4).

The representatives of rows 3 and 6 are explicit: $\text{diag}(1, 1, 4, -1)$ has discriminant $\mathbb{Z}/4$ with $b(g, g) = (\frac{1}{4})^2 \cdot 4 = \frac{1}{4}$; $I_2 \oplus U(2)$ (where $U(2)$ is the Gram $[[0, 2], [2, 0]]$) has discriminant $(\mathbb{Z}/2)^2$ with zero diagonal and $b(x, y) = \frac{1}{2}$; both are odd, of signature (3,1) and $|\det| = 4$ — and Prop1 itself pins each into its row: the triple coincides with exactly one genus. The full list of the 17 putative systems and the merge map $17 \rightarrow 7$ — supplementary **S1** ([\[supplementary/S1_genus_enumeration_17_to_7.md\]](#)(supplementary/S1_genus_enumeration_17_to_7.md)).

(3) Seven triples — and this is counted without symbols. The columns "discriminant group / b / parity" above are read off the symbols and could in principle separate more finely than a genuine isometry of b — in which case "seven distinct triples" would be an artefact of the notation. The separation is therefore taken from an **independent** count from below: on explicit Grams of this scope the triple — with b as an actual finite form, isometry decided by exhaustive search — takes **seven distinct values**, and each is realised. The triple is a genus invariant; there are exactly seven genera and seven values $\implies$ the map "genus $\rightarrow$ triple" is an **injection**. ■

Two checks on the table. Rows 5 and 7 are the pair of §3.4 (diag(1, 1, 2, -2) against $A_1^2 \oplus U$): the same discriminant group, different parity — and **two different** canonical symbols; canonicalisation does not merge what §3.4 separated by an explicit construction. Row 1 is the only genus at $|\det| = 1$, and it is **odd**: the even variant drops out because type II gives oddity $t \equiv 0$, while the oddity formula demands $t \equiv 2$ — this is Remark 2 (§5), re-derived from the existence conditions rather than quoted.

Which invariant separates each pair. The even competitors of the first three ambients are excluded by $\sigma(q) \bmod 8$ (§3.4); at $|\det| = 4$ the groups $\mathbb{Z}/4$ and $\mathbb{Z}/2 \times \mathbb{Z}/2$ are separated by the group structure; the two forms on $\mathbb{Z}/4$ — by the diagonal of b ($[0, \frac{1}{4}, \frac{1}{4}]$ against $[0, \frac{3}{4}, \frac{3}{4}]$); the two odd forms on $\mathbb{Z}/2 \times \mathbb{Z}/2$ — by the count of b -isotropic elements (1 against 3 non-zero); the pair of rows 5/7 — by the parity bit (**T2**), and this is the only place where it carries weight.

§4 · P7.Cor1 — return to preprint-6

P7.Cor1. $\text{disc}(\Lambda_W) = 0 \iff \Gamma$ is Lagrangian $\iff \text{disc}(L) \cong \mathbb{Z}/n$ with $b_L = \langle 1/n \rangle$ — that is, **P6.Thm1(7)**. The condition $\langle 1/n \rangle$ here and everywhere is read **bilinearly** (P7.Def1), in P6's notation: $b_L(g, g) \equiv a/n \pmod{\mathbb{Z}}$ on the generator, a a quadratic residue mod n .

(**Cor1.1**) $\text{disc}(\Lambda_W) \cong \Gamma^\perp/\Gamma \implies \text{disc}(\Lambda_W) = 0 \iff \Gamma^\perp = \Gamma$ (Lagrangian). (**Cor1.2**) Order count:

$$\begin{aligned} |\Gamma| &= |H_L| = |H_t| \quad (\text{i}) \\ |\Gamma|^2 &= |\text{disc}(L)| \cdot n \quad (\text{ii}) \\ |H_L| &\leq |\text{disc}(L)|, \quad |H_t| \leq n \quad (\text{iii}) \\ \implies H_L &= \text{disc}(L), \quad H_t = \text{disc}(\langle -n \rangle), \quad |\text{disc}(L)| = n \quad (\text{iv}) \end{aligned}$$

equality of the products under upper bounds **forces equality of each factor**.

From (iv) it follows not merely that the orders are equal, but that $\text{disc}(L)$ is **isometric** to the cyclic form — this is precisely where the requirement "as a form" starts to work (necessity — §6).

(**Cor1.3**) From (Cor1.2) $H_L = \text{disc}(L)$, so γ is an anti-isometry of **all** of $\text{disc}(L)$ onto $\mathbb{Z}/n$: $b_L(x, y) = -b_t(\gamma x, \gamma y)$. On the generator g we have $\gamma(g) = k \cdot g_t$ with k a unit mod n (γ being an isomorphism), and by (Thm1.5) $b_t(g_t, g_t) = -1/n$, hence

$$b_L(g, g) = -k^2 \cdot b_t(g_t, g_t) = -k^2 \cdot (-1/n) = k^2/n \pmod{\mathbb{Z}}$$

— that is, $b_L = \langle 1/n \rangle$ in the bilinear notation ($a = k^2$ a quadratic residue). The quadratic $q \bmod 2\mathbb{Z}$ does not arise anywhere here — the transfer is purely bilinear. ■

Main argument of the work: preprint-6 is **neither bypassed nor corrected** — it follows from P7.Thm1 by substituting $\text{disc} = 0$. This is a **generalisation that folds back into the printed result**.

§5 · P7.Cor2 — parity rider: the boundary of preprint-6 is exact

P7.Cor2. If the glued overlattice $\Lambda(\Gamma)$ is **even**, then L and $\langle -n \rangle$ are **both** even. **Corollary:** for $p \equiv 1 \pmod{8}$ the condition $\langle 1/n \rangle$ is **not sufficient** — it does not distinguish an even ambient from an odd one.

Proof of necessity. $M = L \oplus \langle -n \rangle \subseteq \Lambda(\Gamma)$, so every vector of M has even norm. ■

The converse does NOT hold: the parity of the overlattice is read off from the norms of the **glue vectors** $\text{mod } 2\mathbb{Z}$, not from the parity of the summands. Witness — our own pair from §3.4: the **even** $M = A_1^3 \oplus \langle -2 \rangle$ glues along one class into the **even** $A_1^2 \oplus U$, and along another into the **odd** $\text{diag}(1, 1, 2, -2)$. This is exactly why parity entered the criterion's data as a separate condition **(T2)**, rather than as a consequence of the summands' parity.

lattice	discriminant form	parity	ambient
$E_8 \oplus \langle 2 \rangle$	$\mathbb{Z}/2, b = \frac{1}{2}$	even	only $II_{9,1}$
$I_8 \oplus \langle 2 \rangle$	$\mathbb{Z}/2, b = \frac{1}{2}$ — the same	odd	$I_{9,1}$

$\implies$ the discriminant form **does not determine** the ambient; an extra **bit** is needed. Hence the boundary of $P6.Thm1(7)$ **IS EXACT**, not conservative.

Remark 2 — the case $p = 3$

$II_{3,1}$ **does not exist**, so for $p = 3$ the even branch is empty, the rider cannot fire, and the condition $\langle 1/n \rangle$ **is sufficient**.

Proof. An even unimodular indefinite lattice exists $\iff sig \equiv 0(\text{mod } 8)$; for $(3,1)$ $sig = 2 \not\equiv 0$. ■ [cited] the classification of indefinite even unimodular [lattices] — the named leg of this remark.

Positively and exactly: the boundary of the condition $\langle 1/n \rangle$ **does not touch** the case $p = 3$, so the printed result of preprint-6 **stands as is**; this work merely outlined its scope from outside.

§6 · P7.Cor3 — the form is necessary

P7.Cor3. The order-only version of the criterion ($|disc(L)| \cdot n = |disc(\Lambda_W)| \cdot index^2$) **is insufficient**.

The ambient is given explicitly, it is 4×4 diagonal **and is itself the witness**:

$\Lambda_W = \text{diag}(1, 1, 2, -2)$, $|disc| = 4$, signature $(3,1)$
 $disc(\Lambda_W) = \mathbb{Z}/2 \times \mathbb{Z}/2$, form $b = \{0, 0, \frac{1}{2}, \frac{1}{2}\}$

t^2	$\backslash \{ \}$	$disc(L) \backslash \{ \}$	$\prime$	$index^2$	gluing arithmetic
−1	4	1	✓		
−2	2	1	✓		
−3	12	9	✓		
−4	16	16	✓		

The gluing arithmetic passes on **ALL four** — and precisely because of that it decides **nothing**. The form distinguishes: $\mathbb{Z}/4$ and $\mathbb{Z}/2 \times \mathbb{Z}/2$ have the same order 4, but **are not isometric**; $disc(\Lambda_W)$ coincides precisely with $\mathbb{Z}/2 \times \mathbb{Z}/2$.

The cost of the omission: without the words "as a form" **P7.Thm1 is false** — it becomes a statement about order and fails at the first non-cyclic discriminant. The corollary that saves the theorem from falsity must say this about itself.

§7 · P7.Thm2 — cost: the law and the closed formula

§7.1 · The law

P7.Thm2(a) — the cost depends ONLY on the bilinear discriminant form of L . If $b_{L_1} \cong b_{L_2}$ (with the same rank and signature), then $\text{cost}(L_1) = \text{cost}(L_2)$.

Corollary (citable form): the cost is a **genus invariant** of L — the genus determines the discriminant form. Invariance by genus is **strictly weaker**: the witness of §7.2(i) has the same cost for **different** genera.

Order of proof: (a) is derived from (b), not from Thm1. This is not a matter of style. After the repair the data of Thm1 contains **parity (T2)**, so the phrase "the criterion is expressed purely in discriminant data" can **no longer** be taken as a premise — it has ceased to be exact. Instead, **the closed formula of §7.3 demonstrably consumes only b :** neither parity nor the Gram matrix enters it. So (a) is proved **after** (b) and **from** (b) — constructively, not rhetorically.

(Thm2.1) The formula of P7.Thm2(b) is expressed via $|\text{disc}(L)|$, $\exp(\text{disc}L)$, and the values of b_L on the elements of $\text{disc}(L)$ — that is, **exclusively** via b as a form; neither the Gram matrix nor parity enters it. Isometric b give the same value of the formula, and the formula equals the cost $\implies \text{cost}$ is a function of b . ■ **(Thm2.2)** The genus determines the signature and the discriminant form $\implies$ on the genus the cost is constant — a **genus invariant**. ■

Why there is no import here. Previously this step relied on "signature + discriminant form is the genus" ([cited] Nikulin). This reference is both **unnecessary** and **too narrow**: only the trivial direction is needed (genus $\implies$ discriminant form), while the converse for **odd** lattices with a **bilinear** form is **false** — our own witness from §5 ($E_8 \oplus \langle 2 \rangle$ and $I_8 \oplus \langle 2 \rangle$: the same bilinear discriminant form, different genera). Nikulin's theorem is formulated for **even** lattices and a **quadratic** form; we do not stretch it beyond its scope (§10).

Why parity (T2) does not affect the cost — and this is visible from the definition, not from hope. *P7.Def5* minimises over **all** ambients, not over ambients of a prescribed parity: every admissible gluing gives **some** integral overlattice (3,1), and it is itself a witness. So the added condition (T2), which distinguishes ambients from one another, **does not shift** the minimum. The witness of §7.2(i) shows this directly: the pair differs precisely in parity and both have cost 1. $\implies$ **the cost half of the work (§7–§8) does not move under the repair of Thm1**, and it has zero imports: neither Eichler, nor genus.

§7.2 · Two different statements — two different witnesses

(i) **The cost is determined by the discriminant form — and it does NOT see either class or genus.** Witness: **the same discriminant form, the same cost, different lattices — even different genera.** Here Gram matrices are mandatory (3×3 , one pair) — the dispute is precisely about the representative:

diag(1,1,6) [[1,0,0],[0,1,0],[0,0,6]] |Aut| 16 ODD disc $\mathbb{Z}/6$, $b(g,g) = 1/6$ cost
1
 $A_2 \oplus \langle 2 \rangle$ [[2,-1,0],[-1,2,0],[0,0,2]] |Aut| 24 EVEN disc $\mathbb{Z}/6$, $b(g,g) = 1/6$ cost
1

$|Aut| 16 \neq 24 \implies$ **not isometric**. Moreover: diag(1,1,6) is **odd** ($e_1^2 = 1$), while $A_2 \oplus \langle 2 \rangle$ is **even** (the entire Gram diagonal is even), and parity is a genus invariant $\implies$ these two lattices lie in **different genera**. The same discriminant form $\implies$ **the same cost** despite this.

The witness proves more than "a genus invariant". The weaker version ("one genus, two classes") would have been a special case of this one. The variable on which the cost depends is the **discriminant form**; the genus would be a superfluous intermediary that, moreover, does not even coincide here.

(ii) **The cost is NOT an invariant of the Bravais TYPE** — the witness is **opposite in form**: the same **type**, **different costs** — $A_2 \oplus \langle 2 \rangle$ (hexagonal) $\rightarrow 1$ versus the row hP (hexagonal) $\rightarrow 3$.

Why (i) does not prove (ii): equal cost for two lattices says that the cost does not see **class** or **genus**, and says nothing about **type**. Only a divergence **within** a type can refute invariance of type.

§7.3 · The closed formula

P7.Thm2(b). $cost(L) = \min |disc(L)| \cdot n_m in(m) / m^2 : m \in D(L)$, with $D(L)$, $n_m in$ as per P7.Def6.

Step 1 — FINITENESS (as a separate and first step: it is precisely this that makes the expression a theorem, not a search)

(Thm2.3) Γ projects into $disc(L)$ injectively (Thm1.3) $\implies \Gamma$ is **isomorphic to a subgroup of** $disc(L)$. (Thm2.4) $\implies m := |\Gamma|$ divides $|disc(L)|$ (Lagrange), and **more precisely** $m | exp(disc L)$, because H_L is cyclic — this is (Thm2.7) below, and it is proved **independently** of this step, so there is no circularity. The divisors are **always** finite in number. (Thm2.5) The set in the definition of $j_m in(m)$ is **non-empty**, and $j_m in(m) \leq m$. Indeed: $m \in D(L)$ guarantees the existence of x of order m ; at $k = 1$ the condition reads as $j \equiv A(mod m)$, where $A := b_L(x, x) \cdot m mod m$ is an integer, because $ord(x) = m$. The class $j mod m$ has a representative in $1, \dots, m$ (at $A = 0$ this is $j = m$) $\implies$ the set is non-empty and the minimum is $\leq m$. Finite. (Thm2.6) There are **infinitely many** admissible pairs (m, n) : for a fixed m the congruence of P7.Def6 admits **the entire residue class** $j mod m$, and with it arbitrarily large $n = m \cdot j$. The finiteness lives not in the pairs, but in the minimisation: the target quantity $|disc(L)| \cdot n / m^2 =$

disc(L)	.j/mstrictly increases in j for fixed m', so the minimum over the class is attained at
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$j_m in(m)$ (non-empty and $\leq m$ — Thm2.5), and m ranges over the finite $D(L)$ (Thm2.4) $\implies$ the minimisation reduces to the finite set $(m, m \cdot j_m in(m)) : m \in D(L)$; the minimum exists, is attained, **without any search bound** D_max . ■

The formula is truncation-stable by construction — not because the search bound $D_m ax$ is taken sufficiently large, but because there **is no bound in it at all** ($D(L)$ from P7.Def6 is the set of divisors of $exp(discL)$, a structural object, not a truncation parameter).

Step 2 — the expression counts exactly what is needed

(Thm2.7) Cyclicity is a consequence, not a convenient special case: $H_t \subseteq \mathbb{Z}/n$ is always cyclic; γ an isomorphism $\implies H_L$ is **forced** to be cyclic. Hence a single m and a single generator suffice **by structure**.

Removes the apparent contradiction with §6: $disc(L)$ can be non-cyclic (that is precisely what §6 is about); what is cyclic is the **glue subgroup**, not the disc.

(Thm2.7') — the scalar congruence of Def6 is equivalent to the FULL anti-isometry. Separately, because $Def6$ operates on a **single number** $b_L(x, x)$, whereas $Def2$ requires equality of the forms on **all** pairs. For cyclic $H_L = \langle x \rangle$, $H_t = \langle y \rangle$ and $\gamma(x) = k \cdot y$, bilinearity gives $b(ax, bx) = ab \cdot b(x, x)$, so

$$b_t(\gamma(ax), \gamma(bx)) = ab \cdot k^2 \cdot b_t(y, y) \text{ and } -b_L(ax, bx) = -ab \cdot b_L(x, x)$$

for all $a, b \iff k^2 \cdot b_t(y, y) = -b_L(x, x)$ — **one** equality on the generators. The value of the form on the generator determines it on the entire cyclic subgroup; hence the scalar condition $Def6$ is no weaker than $Def2$. (Cyclicity is not an assumption, but (Thm2.7).)

Next — the only place with genuine arithmetic: $y = (n/m) \cdot g$ gives $b_t(y, y) = -n/m^2$, and with $n = m \cdot j$ and multiplication by m :

$$j \cdot k^2 \equiv b_L(x, x) \cdot m \pmod{m}$$

hence j ranges over **quadratic shifts by units** k , and j_min is a question about **quadratic residues mod** m .

(Thm2.8) The gluing arithmetic gives $|\det \Lambda_W| = |discL| \cdot n/m^2$; minimisation over (m, n) is precisely the expression.

(Thm2.9) — a candidate from Def6 gives precisely an ambient ADMISSIBLE under Def5. The step is spelled out in full, because $Def5$ minimises **not** over integral overlattices in general, but over those where L is a **primitive timelike section**:

```
admissible gluing (m, j, k)
 $\implies \Gamma = \text{graph}(\gamma), \gamma : H\_L \rightarrow H\_t$  isomorphism (Thm2.7')
 $\implies$  both projections  $\Gamma \rightarrow disc(L), \Gamma \rightarrow disc(\langle -n \rangle)$ 
are injective -- automatically, because it is the graph of an ISOMORPHISM
 $\implies \langle t \rangle$  primitive,  $L$  primitive (Thm1.10)
 $\implies L = t^\perp \cap \Lambda$  : primitivity of  $L$  gives  $L = (L \otimes \mathbb{Q}) \cap \Lambda$ ,
and  $L \otimes \mathbb{Q} = t^\perp \otimes \mathbb{Q}$  (P7.Def4)
 $\implies \Lambda$  is an admissible ambient under P7.Def5
```

Plus Γ is isotropic under P7.Def3 $\implies \Lambda$ is **integral**, of signature (3,1) (Thm1.7, Thm1.8). $\implies$ **the formula = the criterion, no wider:** every member of it is realised, and no realisable ambient falls out from under it. ■

§7.4 · Table of values

Indexed by the **INPUT** of the formula — the discriminant **FORM**, not the Bravais type and not the order of the disc. The table of values must be ordered by what the value depends on. Type is a **label of the chosen representative**. The **group of the disc is NOT an input**. The b column is in the table precisely for this reason: without it the row " $\mathbb{Z}/6 \rightarrow 2$ " would contradict the pair of §7.2 (" $\mathbb{Z}/6 \rightarrow 1$ "), even though both numbers are correct — these are **two different forms on the same group**.

$disc(L)$ group	$\backslash \{ \}$	$disc \backslash \{ \}$	$'$	$b(x, x)$ the gener- ator of the gluing	on cost	witness (n, m)	representative label
0	1	—	1	(1, 1)	cP		
$\mathbb{Z}/2$	2	1/2	1	(2, 2)	tP		
$\mathbb{Z}/4$	4	3/4	3	(12, 4)	cF "FCC"		
$\mathbb{Z}/6$	6	1/3	2	(3, 3)	oP		
$\mathbb{Z}/8$	8	1/2	4	(8, 4)	tI		
$\mathbb{Z}/3 \times \mathbb{Z}/3$	9	1/3	3	(3, 3)	hP		
$\mathbb{Z}/4 \times \mathbb{Z}/4$	16	1/2	8	(8, 4)	cI "BCC"		
$\mathbb{Z}/2 \times \mathbb{Z}/8$	16	3/8	6	(24, 8)	oC		
$\mathbb{Z}/2 \times \mathbb{Z}/10$	20	2/5	8	(10, 5)	hR		
$\mathbb{Z}/24$	24	1/8	3	(8, 8)	oF		
$\mathbb{Z}/2 \times \mathbb{Z}/14$	28	3/14	6	(42, 14)	mC		
$\mathbb{Z}/51$	51	1/17	3	(17, 17)	aP		
$\mathbb{Z}/69$	69	2/69	2	(138, 69)	mP		
$\mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/2$	96	5/24	20	(120, 24)	oI		
$\mathbb{Z}/24$							

How to read the b column. This is the value on the generator of the **glue subgroup** (an element of order m) — it is exactly this that enters the congruence $j \cdot k^2 \equiv b \cdot m \pmod{m}$. The value on the generator of the **whole** disc can be different: for the row oP $b(g, g) = 5/6$ on the generator of $\mathbb{Z}/6$, whereas the table shows 1/3 — the value on $x = 2g$ of order 3. **Every displayed witness can be checked from its row:** $j = n/m$, $cost = |disc| \cdot n/m^2$; for **non-cyclic** forms the independent check of **minimality** requires the full discriminant form — supplementary **S2** (`[supplementary/S2discriminant_form_grams.md]`(`supplementary/S2_discriminant_form_grams.md`)). **Two forms on $\mathbb{Z}/6$ that justify this column:** $b(g, g) = 5/6$ (row oP) gives cost 2; $b(g, g) = 1/6$ (the pair of §7.2) gives cost 1. One group, different costs — because the forms differ.

(The table — values of the formula, by a single method; representatives validated by $|Aut|$, 14/14; Gram matrices — supplementary.) For **non-cyclic** $disc(L)$ the b column gives the value only on the generator of the gluing, not the whole form: the row verifies the **upper bound** of the cost, while minimality is with respect to the full discriminant form; the full Gram matrices of the discriminant forms of all six non-cyclic rows — supplementary **S2** (`[supplementary/S2discriminant_form_grams.md]`(`supplementary/S2_discriminant_form_grams.md`)).

Two things become visible that a table by types was hiding: (i) at the same order the form distinguishes: $\mathbb{Z}/4 \times \mathbb{Z}/4 \rightarrow 8$ versus $\mathbb{Z}/2 \times \mathbb{Z}/8 \rightarrow 6$; (ii) **the cost is NOT monotone in $|disc|$** — $\mathbb{Z}/69 \rightarrow 2$, $\mathbb{Z}/4 \rightarrow 3$, $\mathbb{Z}/8 \rightarrow 4$: a larger disc can be **cheaper**. $\implies$ **no "size leads" whatsoever**; the cost is a function of the discriminant form, and only that.

§8 · P7.Cor4 — centring addresses

P7.Cor4. FCC — $cost = 3$, **BCC** — $cost = 8$; both unattainable on the unimodular stratum, but with **different** addresses, and the difference is explained by the **discriminant form**.

Table 1 — the blocking mechanism: three lines, not one

For FCC $disc = \mathbb{Z}/4$, $b = 3/4$; the admissible $m \in 1, 2, 4$, and $d = |disc| \cdot j/m$ with $j \cdot k^2 \equiv b \cdot m(modm)$.

'd = \{\}	det $\Lambda_W \setminus \{\}$	'	status	what exactly blocks it
d = 1	blocked	requires $k^2 \equiv 3(mod4)$, but squares mod 4 are 0, 1: quadratic residue		
d = 2	blocked	both routes fall on congruences modulo 2 : $m = 2$ gives $1 \not\equiv 0(mod2)$, $m = 4$ gives $2k^2 \equiv 3(mod4)$ — an even number against an odd one		
d = 3	blocked by nothing	it is the minimum itself — the over-lattice is explicit, $\setminus \{\}$	det \{\}	$= 3, (3, 1), (n, m) = (12, 4)'$

On the word "parity" in the row $d = 2$. Here it is the **parity of integers** in the congruence at issue, **not** the parity of the lattice (type I/II) from condition **(T2)**. The homonym is named: after §3 "parity" in this work is a load-bearing word, and in Table 1 it means **something else**. **"Form** blocks, not order" — a conclusion from three lines, not a substitute for them.

The status of the number 8 for BCC — so that two levels are not read as one. $cost(BCC) = 8$ is obtained **abstractly**: the formula P7.Thm2(b) computes the cost from the b of L itself, and the realisability of each term is automatic (Thm2.9). It **does not depend** on the open item of §11 — that concerns the **full isometry** $\Gamma^\perp/\Gamma$ for BCC, i.e. the **identification of the specific ambient**, not the value of the cost. The cost answers "how expensive", the identification "which one exactly"; the latter remains open.

This is the **native reason** for excluding FCC from the unimodular stratum: earlier it stood as an **observed obstruction**, now it is **derived** from the criterion. The **asymmetry** $3 < 8$ is explained by **different forms** — and **not** by monotonicity in size (§7.4).

§9 · Measured alongside: *cost* and n_{min} — not one scale

Is the "ambient cost" not another expression for n_{min} (P6)? **No:**

`Spearman( n_min , cost )  $\approx$  0.78` on one set of representatives

`Spearman( n_min , cost )  $\approx$  0.36` on an alternative set

The coefficient depends on the chosen set of representatives and is not cited as a value.

Conclusion — as two separate statements, each with its own witness (one is not proved without the other: two order reversals **do not** refute a functional dependence — a function can perfectly well have $f(43) = 3$, $f(13) = 8$):

(i) **The order is not consistent (non-monotonicity):**

triclinic	<code>n_min = 43 , cost = 3</code>
rhombohedral	<code>n_min = 13 , cost = 8</code>

a larger n_{min} — a smaller cost; it is exactly this (and only this) that *Spearman* < 1 reflects.

(ii) ***cost* is not a function of n_{min} — a collision:** n_{min} is an invariant of **holohedry** (P6), while the cost is not: both lattices of the pair in §7.2(ii) are hexagonal, i.e. they have **the same** holohedry and $n_{min} = 6$, but costs **1** ($A_2 \oplus \langle 2 \rangle$) and **3** (the *hP* row). The same argument, different values — there is no function.

§10 · Relation to prior work and sources

mechanism	source
the terminology of gluing (glue group / glue vectors); the discriminant-form theory of the even case (the identity $disc(\Lambda) \cong \Gamma^\perp / \Gamma$ for the odd case is proved in this work — P7.Lem1)	V. V. Nikulin , <i>Integral symmetric bilinear forms and some of their applications</i> , Izv. Akad. Nauk SSSR 43 (1979); English transl. Math. USSR Izv. 14 (1980) — §1.5, §1.5.1. J. H. Conway, N. J. A. Sloane , <i>Sphere Packings, Lattices and Groups</i> , 3rd ed., Springer 1999 — Ch. 4
class = spinor genus for indefinite forms of rank ≥ 3	M. Eichler , <i>Quadratische Formen und orthogonale Gruppen</i> , Springer 1952. M. Kneser , <i>Klassenzahlen indefiniter quadratischer Formen...</i> , Arch. Math. 7 (1956). Modern exposition: J. W. S. Cassels , <i>Rational Quadratic Forms</i> , Academic Press 1978 — Ch. 11 ; O. T. O'Meara , <i>Introduction to Quadratic Forms</i> , Springer — §102
one spinor genus in the genus for small $\backslash\{\}$ correctness of the Conway–Sloane calculus for 2-adic lattices (the first published proof; §6 — the correction of the false CS canonical form , following Allcock, Memoirs AMS 220; it carries the merging of §3.5: Lemma 5.2, Lemma 6.1, Theorem 6.2)	$\det\backslash\{\}$ D. Allcock, I. Gal, A. Mark , <i>The Conway–Sloane calculus for 2-adic lattices</i> , arXiv : 1511.04614

mechanism	source
indefinite even unimodular ($sig \equiv 0 \bmod 8$); classification of the odd ones	J.-P. Serre , <i>A Course in Arithmetic</i> , GTM 7 — Ch. V . J. Milnor , D. Husemoller , <i>Symmetric Bilinear Forms</i> , Springer 1973 — Ch. II §5
the Milgram formula (σ of the discriminant form $\bmod 8$); exclusion of the even competitor	Milnor–Husemoller , <i>Symmetric Bilinear Forms</i> — Appendix ; SPLAG — formula (30)
existence conditions for p-adic symbols — the load-bearing step of P7.Prop1 (§3.5); the "canonical form" of §7.6 is not used in the steps: its uniqueness is false in general (see the AGM row)	Conway–Sloane , SPLAG 3rd ed. — Ch. 15, §7.7 (Theorems 10, 11)
genus symbols (for reference; Nikulin’s existence theorems §1.9/§1.11 — the even case, not used in the steps of this work: all ambi-ents and overlattices are built explicitly)	Nikulin §1.9, §1.11
the crystallographic restriction in rank 3 (<i>historical note</i>)	19th-century classic; modern expositions — W. Scherrer (1946), H. S. M. Coxeter , <i>Introduction to Geometry</i> (1969)

THE SCOPE LINE, WITHOUT WHICH THE REFERENCE TO NIKULIN READS WIDER THAN IT IS. Nikulin’s discriminant-form theory is formulated for **EVEN** lattices. **Our case is odd** — and it works **natively**, because the whole criterion is built from the outset on the **bilinear** discriminant form $\bmod \mathbb{Z}$ (*P7.Def1*), not on the quadratic $\bmod 2\mathbb{Z}$, which is defined only for even lattices. $\implies$ we do **not** apply Nikulin’s result outside its scope: from there we take the **terminology of gluing and the identity** $disc(\Lambda) \cong \Gamma^\perp/\Gamma$, while the odd case carries its own construction. The price of this choice — that the bilinear form does not see parity — is paid explicitly: the bit sits in the data of the criterion (**T2**), and is not hidden in the import.

Theorem numbers are given only where checked against the edition; otherwise — the chapter or section.

What is genuinely this work’s own:

1. **the criterion relative to a fixed ambient** with $\Gamma^\perp/\Gamma \cong disc(\Lambda_W)$ as a form and with **parity** in the data (the odd case);
2. **folding back**: $disc = 0$ reproduces the $\langle 1/n \rangle$ condition of preprint-6;
3. **the cost as an invariant of the discriminant form** plus a **closed formula** with a finite range — **without a search parameter**;
4. **the addresses of the centring**s with the blocking mechanism separated out (Table 1).

§11 · Limits and the open

(a) Open

question	status
characterisation of (T0)-lattices — for which Λ_W the triple "signature + b + parity" determines the genus in general form ~~the lemma "no even competitor"~~	open for arbitrary det and rank $\neq 4$; for rank 4, sig (3,1), '\{\}
the full form of $\Gamma^\perp/\Gamma$ for BCC	closed — written out as the table of §3.4 ($\sigma \bmod 8$, all refinements of q , the enumeration is complete) the order and factors are computed; the full isometry has not been done. This does not concern the cost — $\text{cost}(BCC) = 8$ is abstract (§8)
γ-freedom for a non-cyclic disc a simpler form of $j_{\min}(m)$ the cost as an invariant of TYPE	open without a source possibly via quadratic residues / CRT closed negatively — witness §7.2(ii): the same type, different costs

A bridging line (because $A_2 \oplus \langle 2 \rangle$ works twice): **there, the same discriminant form — the same cost even with different genera; here, the same TYPE — a different cost. This is exactly what shows that the discriminant form is the variable.**

(b) Parking, said out loud

The sample $n \leq 4$ does not even reproduce the **known** unimodular image (it gives *tetragonal, cubic* instead of seven; *triclinic* requires $n = 43$) $\implies$ a "by genus" difference on it would be a **truncation artefact**. Fincke–Pohst up to $n = 43$ with a budget is needed.

(c) On method: why the theorem has no search bound

A bounded search (limit $D_{\max} = 40$) **gave inflated cost values — three times**. The reason is each time the same: for large discriminants the optimal m is large, $n = d \cdot m^2 / |\text{disc}|$ jumps past the limit, the row **drops out of the enumeration**. (*Notation: $D_{\max}$ is the truncation limit of the old search; it has nothing in common with the set of divisors $D(L)$ in P7.Def6.*)

The finiteness step (Thm2.3–Thm2.6) exists precisely so that the theorem has no search bound at all. $m|\exp(\text{disc}L)$ is a **structural** condition, not a computational one. **Conclusion of the method: a named structure is not verified by scanning** — if a problem is finite by structure, this must be **used**, not approximated by a bound.

(d) What is not our result

The window (quasiperiodicity) is a separate input, not a consequence of the criterion. Finite-order integer isometries of $I_{3,1}$ have orders from 1, 2, 3, 4, 6: axes of orders **5, 8, 12** do not preserve (3,1) — **measured**; order **10** is excluded as a consequence (its square has order 5) — **derived**. A detailed treatment is not part of this work.

This is a statement about **scope**, not a promise: it says **what is absent**, without "later we will show".